We will spend the first part of this course fleshing out this picture, in which we are particularly interested in understanding the integral closure of the rings $\mathbb{Z}$ and $\mathbb{F}_q[t]$ in finite extensions of their fraction fields (such integral closures are known as *rings of integers*), and the prime ideals of these rings. Where possible we will treat number fields and function fields on an equal footing, but we will also note some key differences. Surprisingly, the apparently more complicated function field setting often turns out to be simpler than the number field setting; for example, the analog of the Riemann hypothesis in the function field setting (the Riemann hypothesis for curves), is not an open problem. It was proved by André Weil in the 1940s [5]; a further generalization to varieties of arbitrary dimension was proved by Pierre Deligne in the 1970s [3].

Zeta functions provide the tool we need to understand the distribution of primes, both in general, and within particular residue classes; the proofs of the prime number theorem and Dirichlet's theorem on primes in arithmetic progressions both use zeta functions in an essential way. Dirichlet's theorem states that for each integer $m > 1$ and each integer a coprime to m, there are infinitely many primes $p \equiv a \bmod m$. In fact, more is true: the Chebotarev density theorem tells us that for each modulus m the primes are equidistributed among the residue classes of the integers a coprime to m. We will see this and several other applications of the Chebotarev density theorem in the later part of the course.

Before we begin, let us note the following.

Remark 1.1. Our rings always have a multiplicative identity that is preserved by ring homomorphisms (so the zero ring in which $1 = 0$ is not an initial object in the category of rings, but it is the terminal object in this category). Except where noted otherwise, the rings we shall consider are all commutative.

1.2 Absolute values

We begin with the general notion of an absolute value on a field; a reference for much of this material is [4, Chapter 1].

Definition 1.2. An *absolute value* on a field k is a map $|\ |\colon k \to \mathbb{R}_{\geq 0}$ such that for all $x, y \in k$ the following hold:

1. $|x| = 0$ if and only if $x = 0$;
2. $|xy| = |x||y|$;
3. $|x + y| \leq |x| + |y|$.

If the stronger condition

4. $|x + y| \leq \max(|x|, |y|)$

also holds, then the absolute value is *nonarchimedean*; otherwise it is *archimedean*.

Example 1.3. The map $|\ |\colon k \to \mathbb{R}_{\geq 0}$ defined by

$$|x| = \begin{cases} 1 & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

is the *trivial absolute value* on k. It is nonarchimedean.